Journal of Physics:
Conference Series

PURPOSE-LED PUBLISHING™

PAPER • **OPEN ACCESS**

MEMS-IMU automatic calibration system design and implementation

To cite this article: Jingxiao Wang and Ning Liu 2023 *J. Phys.: Conf. Ser.* **2492** 012005

View the article online for updates and enhancements.

You may also like

- An *in situ* hand calibration method using a pseudo-observation scheme for low-end inertial measurement units
You Li, Xiaoji Niu, Quan Zhang et al.

- FDO-Calibr: visual-aided IMU calibration based on frequency-domain optimization
Yunfei Liu, Zhitian Li, Linhui Xiao et al.

- Dual-antenna GNSS-aided robust MEMS IMU initial alignment under the earth-centered-earth-fixed frame
Siyu Zhang, Weiming Tang, Chenlong Deng et al.

This content was downloaded from IP address 193.40.12.10 on 28/07/2025 at 12:58

MEMS-IMU automatic calibration system design and implementation

Jingxiao Wang, Ning Liu *

Beijing Information Science and Technology University, Technology Beijing Key Laboratory of High Dynamic Navigation Technology, Beijing 100192, China

* Corresponding author: ning.liu@bistu.edu.cn

ABSTRACT. Aiming at the problems of traditional calibration, long calibration cycles, and low efficiency of traditional error calibration methods for MEMS inertial devices, a MEMS IMU automatic calibration system is proposed. Firstly, the error mechanism of MEMS-IMU is analyzed, the error model is established, and the calibration scheme is designed. Then, the automatic control of the turntable and the automatic data acquisition of MEMS inertial devices are realized by designing the data interaction mechanism between the system equipment. Finally, the least square method is used to identify the model parameters. The experimental results show that this method can meet the requirements of multi-position and multi-rate calibration. After compensation, the relative errors of the MEMS gyroscope and accelerometer are 0.04% and 0.15%, respectively, and the cross-coupling error of the MEMS gyroscope is reduced to about 0.1 °/s. At the same time, the whole calibration process is easy to operate, which greatly shortens the calibration time and improves the calibration efficiency.

1. INTRODUCTION

The Micro Electro Mechanical System Inertial Measurement Unit (MEMS-IMU), whose main components are MEMS accelerometers and MEMS gyroscopes, is a device for sensitizing carriers to their linear acceleration and angular velocity, is now widely used in various fields[1][2][3]. MEMS inertial sensors without error compensation are subject to measurement errors when they are excited by error sources[4], which affects their output accuracy in use. SYED[5] propose a "rotating unit" turntable designed to provide angular velocity to the gyroscope to identify and compensate for the main sensor errors. LI[6] compensates for the error by using the trained neural network, but the network training requires a lot of pre-learning and constantly updates the model when it is used, which is not suitable for practical engineering due to a large amount of calculation.

However, no matter which calibration method is used, the current calibration still depends on the manual operation of the tester. This traditional method requires the tester to operate the turntable controller manually, manually collect and store data from the MEMS sensor, and finally process the acquired data to obtain the calibration parameters of the MEMS sensor[7]. This method not only has low calibration efficiency but also consumes a lot of labor and material resources. YIN[8] designed an automatic test system for MIMU, which is capable of satisfying error compensation for gyroscopes and accelerometers. Wang[9] designed an automatic calibration system based on temperature changes. LIU[10] designed a system-level automatic test calibration method for MEMS-IMU, but the filtering algorithm is complex and computationally intensive.

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

To solve the above problems, this paper proposes a MEMS-IMU automatic calibration system, which realizes turntable control, curve drawing, data acquisition, error compensation, and other functions through various devices in the system. The experimental equipment in this paper adopts a turntable with a three-axis speed position mode. According to the error characteristics, the error models of the MEMS gyroscope and accelerometer are established, respectively. The least square method can quickly identify the parameters of the error model. At the same time, this paper also proposes an information interaction scheduling mechanism in the system to optimize the sequence logic between functions, automate the calibration process, and finally realize the rapid automatic calibration of MEMS-IMU.

2. MEMS-IMU error model

2.1 Establishment of MEMS-IMU error model

The main types of error in MEMS gyroscopes are scale factor error, zero bias error, cross-coupling error, and output non-linearity when the MEMS gyroscope is at a large range of speeds[11][12]. Based on the above analysis of MEMS gyroscope errors, the error model can be divided into the following three parts: scale variation calibration, linearity calibration, and cross-coupling calibration. The error models can be established as:

$$\begin{bmatrix} \tilde{\omega}_x \\ \tilde{\omega}_y \\ \tilde{\omega}_z \end{bmatrix} = \begin{bmatrix} k_{\omega x} & & \\ & k_{\omega y} & \\ & & k_{\omega z} \end{bmatrix} \left(\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} - \begin{bmatrix} \sigma_{\omega x} \\ \sigma_{\omega y} \\ \sigma_{\omega z} \end{bmatrix} \right) \tag{1}$$

where $\tilde{\omega}_x$ (i=x,y,z) is scale variation calibration output values of MEMS gyroscope; ω_x (i=x,y,z)is the raw output value of MEMS gyroscope; $\sigma_{\omega x}$ (i=x,y,z)is zero bias of MEMS gyroscope; $k_{\omega x}$ (i=x,y,z)is the scale factor of MEMS gyroscope.

$$\begin{bmatrix} \bar{\omega}_x \\ \bar{\omega}_y \\ \bar{\omega}_z \end{bmatrix} = \begin{bmatrix} \tilde{\omega}_x & & \\ & \tilde{\omega}_y & \\ & & \tilde{\omega}_z \end{bmatrix}^2 \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} + \begin{bmatrix} \tilde{\omega}_x & & \\ & \tilde{\omega}_y & \\ & & \tilde{\omega}_z \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} + \begin{bmatrix} c_x \\ c_y \\ c_z \end{bmatrix} \tag{2}$$

where $\bar{\omega}_i$ (i=x,y,z) is the linearity calibration output values of MEMS gyroscope; j_i (i=x,y,z; j=a,b,c)is the MEMS gyroscope linearity fitting factor.

$$\begin{bmatrix} \hat{\omega}_x \\ \hat{\omega}_y \\ \hat{\omega}_z \end{bmatrix} = \begin{bmatrix} 1 & M_{xy} & M_{xz} \\ M_{yx} & 1 & M_{yz} \\ M_{zx} & M_{zy} & 1 \end{bmatrix} \begin{bmatrix} \bar{\omega}_x \\ \bar{\omega}_y \\ \bar{\omega}_z \end{bmatrix} \tag{3}$$

where $\hat{\omega}_i$ (i=x,y,z) is the cross-coupling calibration output values for MEMS gyroscope; M_{ij} (i,j=x,y,z)is the cross-coupling coefficient of MEMS gyroscope.

The error model of MEMS accelerometers is similar to that of gyroscopes, and since this paper is an experiment conducted on a three-axis turntable under a gravitational field of 1g, only the accelerometers are varied in magnitude, and the model can be simplified as:

$$\begin{bmatrix} \tilde{a}_x \\ \tilde{a}_y \\ \tilde{a}_z \end{bmatrix} = \begin{bmatrix} k_{ax} & & \\ & k_{ay} & \\ & & k_{az} \end{bmatrix} \left(\begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} - \begin{bmatrix} \sigma_{ax} \\ \sigma_{ay} \\ \sigma_{az} \end{bmatrix} \right) \tag{4}$$

where $\tilde{a}_i$ (i=x,y,z) is the scale variation calibration output values of MEMS accelerometer; a_i (i=x,y,z)is the raw output value of MEMS accelerometer; σ_{ai} (i=x,y,z)is zero bias of MEMS accelerometer; k_{ai} (i=x,y,z)is the scale factor of MEMS accelerometer.

3. Automatic calibration system design

3.1 General design of the system

The automatic calibration system mainly consists of the upper computer calibration software, motion control board, and the three-axis turntable (the turntable controller and the turntable are considered as one). As shown in Figure 1, the automatic calibration system framework.

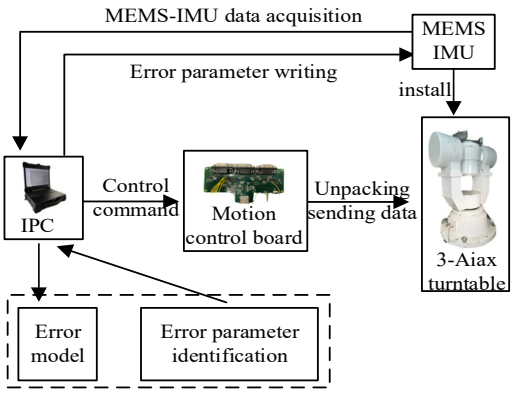

Figure 1 Automatic calibration system framework

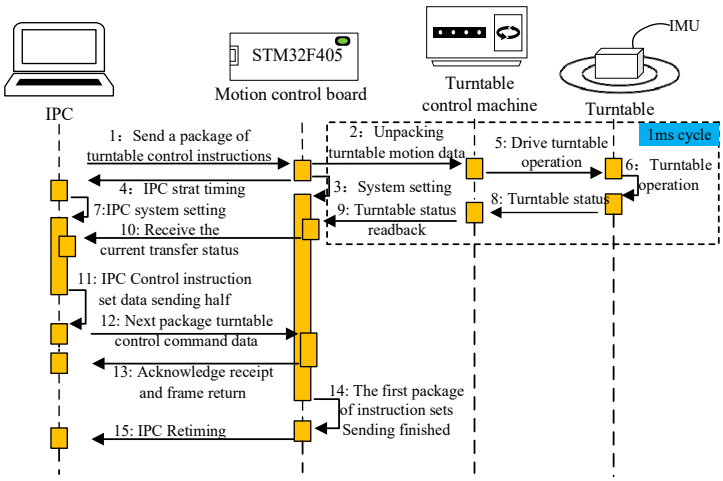

Figure 2 Data scheduling design of system equipment

Theoretically, the upper computer needs to send the data to the turntable with a period of 1ms. Still, due to the large timer error of the Windows system[13], if the turntable receives two frames of motion data from the upper computer with too long an interval, it will cause the turntable controller not to process the motion data. Thus the turntable will not continue to work. Therefore, the data scheduling mechanism between the system devices, as shown in Figure 2, is designed to achieve a precise timing of 1ms with the help of the motion control board. The motion control board hosts a packet (100 ms) of turntable motion commands from the host computer and sends them to the turntable controller at a timed interval of 1 ms. After 50 ms the upper computer sends the next packet (100 ms) of turntable motion data to the board to compensate for the loss of data caused by the Windows system timer error. This enables precise control of the turntable's motion by the host computer.

3.2 Modular functions of the system

The modular functions of the automatic calibration system are:

1) friendly human-machine interface; 2) turntable control function; 3) data communication function; 4) calculation of calibration parameters; 5) dynamic curve plotting of turntables and MEMS-IMU.

4. Experimental validation

4.1 Experimental design

According to the method given in this paper, a high-precision rate position three-axis turntable was used, the MEMS-IMU developed in the laboratory was selected, and the calibration environment is shown in Figure 3. The three-axis turntable is controlled by the automatic calibration system designed above, and the data of MEMS-IMU is collected and stored at the same time. During the calibration process, the MEMS-IMU will rotate at different rates and positions, the multi-rate experiment is used to calibrate the error parameters of the MEMS gyroscope; The multi-position experiment is used to calibrate the error parameters of the MEMS accelerometer. When the turntable rotates to each position point, it collects accelerometer and gyroscope data statically and then rotates to each rating point to collect gyroscope data. After the rate point calibration, the turntable reaches the next position through corresponding rotation to calibrate other axes. The specific steps of the calibration experiment are as follows:

1) Fix the tooling of the MEMS-IMU to be tested on the inner frame of the three-axis turntable, and establish communication with the equipment to be tested and the turntable in the calibration software.

2) Read the calibration file in the calibration software, generate the turntable motion command, and send it to the turntable controller through the motion control board at a time of 1 ms, which drives the turntable to run.

3) The turntable rotates according to the path designed in Table 1. In Table 1, the axis represents the sensitive axis of the gyroscope. Excites the three sensitive axes of the MEMS gyroscope and accelerometer in turn through different positions and rates, and collects the data of the device to be measured at the same time.

4) After the turntable runs, the collected data is processed. According to the error model built in Section 1, the error calibration parameters are quickly fitted using the least square method, and the calibration parameters are written into the device to be measured. At the same time, the error parameters are displayed on the software interface panel.

Table 1 MEMS-IMU position and turntable speed

Position	Accelerometer(m/s^2)			Gyroscope	
	X	Y	Z	Axis	Turntable excitation value (°/s)
①	0	0	9.8	Z	±1000、±500、±200、0
②	9.8	0	0	X	±100、±50、±10、0
③	0	9.8	0	Y	±100、±50、±10、0
④	0	0	-9.8	Z	±1000、±500、±200、0
⑤	-9.8	0	0	X	±100、±50、±10、0
⑥	0	-9.8	0	Y	±100、±50、±10、0

Table 2 Effect after MEMS accelerometer calibration

Axis	Ideal value (m/s^2)	Compensation value (m/s^2)	Relative error (%)
X	9.8	9.812994	0.1326
	-9.8	-9.792356	0.0748
Y	9.8	9.783898	0.1643

Z	-9.8	-9.821736	0.2218
	9.8	9.819717	0.2012
	-9.8	-9.813984	0.1427

4.2 Experimental calibration results

Taking the MEMS gyroscope Z-axis as an example, Figure 4 shows the comparison before and after the change of the MEMS gyroscope scale, which shows that the MEMS gyroscope output is converted from the digital amount of the raw electrical signal at the beginning to the actual analog amount in degrees per second (°/s).

Figure 3 Calibration environment

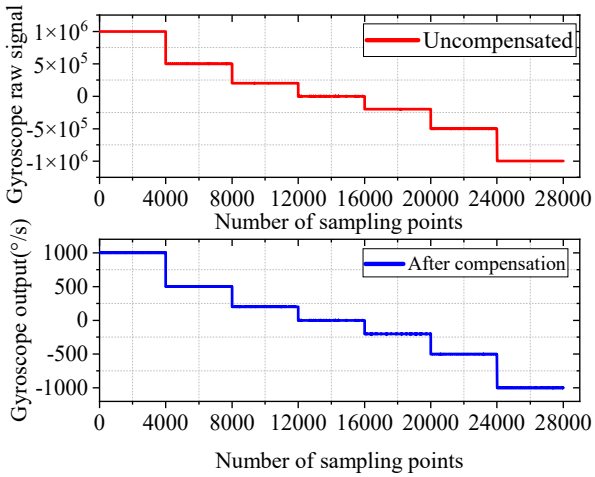

Figure 4 Scale variation calibration before and after comparison

The average relative error of each axis after linearity compensation is 0.04%. Figure 5 shows the MEMS gyroscope before and after cross-coupling compensation. Due to the installation process of the MEMS gyroscope, the non-sensitive axis output will be cross-coupled when rotating along the sensitive axis. The average output of the non-sensitive axis after compensation is 0.1066°/s (X-axis) and 0.0931°/s (Y-axis), respectively, which shows that the cross-coupling error can be effectively suppressed.

The MEMS accelerometer scale changes the same as the gyroscope, converting from the raw electrical signal to m/s^2. Table 2 shows the relative error of the MEMS accelerometer at each position after compensation; the average relative error of each axis after compensation is 0.1562%.

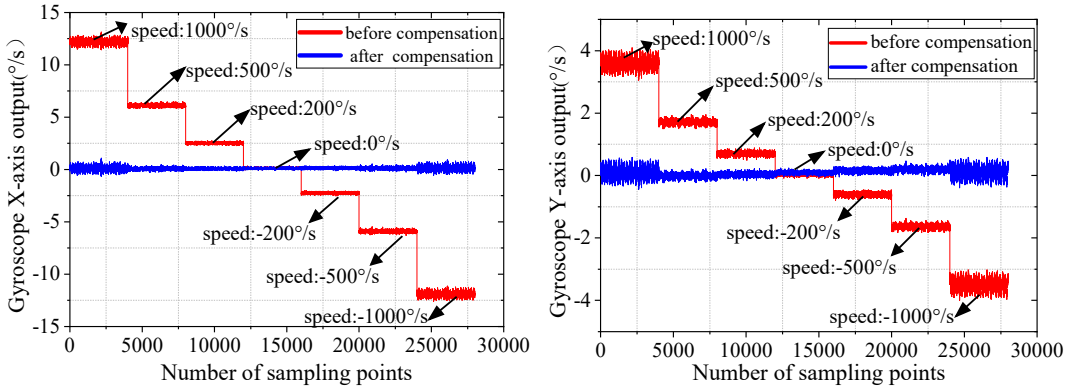

Figure 5 Cross-coupling calibration before and after comparison

5. CONCLUSION

Before the formal use of MEMS-IMU, its error source needs to be calibrated and compensated to ensure the accuracy and reliability of the output data. In this paper, the MEMS-IMU automatic calibration system is designed to solve the problems of the complex, time-consuming, and inefficient calibration process, so that the turntable operation and equipment data acquisition can be fully automated. The experimental results show that the system can effectively compensate for MEMS-IMU error and has certain engineering value.

References

[1] Shaeffer, D. K. MEMS inertial sensors: A tutorial overview[J]. IEEE Communications Magazine, 2013, 51(4): 100-109.
[2] Miguel, J., Lechuga, Y., Martinez, M. AFM-Based Characterization Method of Capacitive MEMS Pressure Sensors for Cardiological Applications[J]. Micromachines, 2018,9(7): 342.
[3] Zhu, J. X., Liu, X. M., Shi, Q. F., et al. Development Trends and Perspectives of Future Sensors and MEMS/NEMS[J]. Micromachines, 2020,11(1): 7.
[4] Fang, B., Chou, W. S., Ding, L. An Optimal Calibration Method for a MEMS Inertial Measurement Unit Regular Paper[J]. International Journal of Advanced Robotic Systems, 2014,11(1): 314-319.
[5] Syed, Z. F., Aggarwal, P., Goodall, C., et al. A new multi-position calibration method for MEMS inertial navigation systems[J]. Measurement science & technology, 2007,18(7): 1897-1907.
[6] Li, R. B., Yan, J. S. S., Liu, G., et al. MEMS-IMU error model and calibration method based on LSTM deep neural network[J]. Journal of Chinese Inertial Technology, 2020, 28(02): 165-171.
[7] Sun, J., Zou, J., Hu, T. A Fast Calibration Method for MEMS Inertial Sensors Based on 24-Position[J]. Piezoelectrics & Acousto-optic, 2019, 41(3): 440-444.
[8] Yin, J., Zhu, X., Wu, Z. Design of an automated batch calibrating system for MIMU[C]//2016 Tenth International Conference on Sensing Technology(ICST), IEEE, 2016: 336-345.
[9] Wang, Y., Dou, L., He, W., et al. Design of Batch Automatic Calibration System for MEMS Inertial Attitude Module[J]. Electronic Measurement Technology, 2020, 43(7): 161-166.
[10] Liu, Z. W., Cui, M., Zhang, P., et al. Rapid Calibration Method of MIMU 24 Error Parameters Based on LABVIEW[J]. Piezoelectric & Acousto-optic, 2020,42(06): 854-858.
[11] Tang, Q. J., Wang, X. J., Yang, Q. P. Scale factor model analysis of MEMS gyroscopes[J]. Microsystem Technologies, 2017,23(5): 1215-1219.
[12] Zhao, R. J., Hu, B. Q., Tian J. Y., et al. Research on practical calibration compensation method of MIMU[J]. Journal of physics. Conference series, 2020,1550(3): 32020.
[13] Ji, S. T., Qin Y. Y., Liu, Z. R. Research on Automation Test System and Software of Fiber-Optic Gyro[J]. Measurement & Control Technology, 2012, 31(5): 77-80.